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Short communication: Accuracy of estimation of lameness, injury, and cleanliness prevalence by dairy farmers and veterinarians

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ABSTRACT

Lameness, injuries, and cleanliness are considered important indicators of dairy cow welfare, milk production, and milk quality. Previous research has identified that farmers globally underestimate the prevalence of these cow-based measurements, but no information on the perceptions of veterinarians is available. Because veterinarians are often perceived as the main providers of health advice on farms, the objective of the present study was to evaluate the relationship between the true prevalence of lameness, injury (hock, knee, neck), and cleanliness (udder, legs, flanks), and the estimated prevalence of these issues by farmers and their herd veterinarians. A cross-sectional study was conducted between February 2016 and July 2017. First, the farm owner and the herd veterinarian were asked to estimate the prevalence of lameness, of neck, knee and hock injuries, and of udder, leg, and flank cleanliness on the farm. The research team then visited the farm and scored all lactating cows in the herd for each measurement. Linear regression models were used to assess the relationship between the prevalence estimated by the veterinarians and the farmers, of each cow-based measurement, and the true prevalence on the farm. The 93 herds enrolled had a median of 55 milking cows and were housed in tiestall (90%) and freestall (10%) barns. Ten herd veterinarians participated and were involved with 2 to 22 enrolled farms each. A wide variation was detected in the true prevalence of the different cow-based measurements among herds (lameness: range = 19–72%, median = 36%; neck injuries: range = 0–65%, median = 14%; knee injuries: range = 0–44%, median = 12%; hock injuries: range = 0–57%, median = 25%; dirty udder: range = 0–55%, median 13%; dirty legs: range = 0–91%, median = 18%; and dirty flanks: range = 0–82%, median = 20%). For both veterinarians and farmers, the perception of each cow-based measurement prevalence increased incrementally as the herd's true prevalence increased. Overall, farmers and veterinarians underestimated cow-based measurements. Farmers and veterinarians more accurately estimated lameness prevalence in herds with higher prevalence than in herds with low prevalence, suggesting a better awareness of the issue on farms with lameness problems. Injuries were less accurately estimated in herds with higher injury prevalence compared with herds with lower prevalence, suggesting an opportunity for better knowledge transfer in this area.
Key words: agreement, prevalence, perception, welfare

Short Communication

Lameness in dairy cattle has been identified by multiple stakeholders as a major welfare and economic concern (Huxley, 2012; von Keyserlingk et al., 2013). Herd prevalence of lameness, including mild to severe cases, has been reported to be between 25 and 55% on average (Barker et al., 2010; von Keyserlingk et al., 2012; Bran et al., 2018). In 2015, a national cross-sectional study found a Canadian lameness prevalence of 25%, with a regional difference of over 20% higher prevalence in Québec compared with other provinces (Croyle, 2020). Hypotheses for this higher prevalence in Québec included not using professional hoof trimmers, the type of housing, and the housing design. Although the reason for this difference was not clear, it became clear that lameness should be addressed as a major issue in this province, and measures should be taken to decrease its prevalence.

Studies have shown that farmers underestimate the prevalence of lameness in their own herds (Espejo et al., 2006; Leach et al., 2010a; Higginson Cutler et al., 2017), which could explain their inaction to reduce lameness on their farms. Farmers also mentioned that lameness was not a big problem, or not a priority, as potential barriers for putting efforts into controlling lameness (Leach et al., 2010a). Moreover, most farmers were motivated to control lameness, as it resulted in the pride of having a healthy herd (Leach et al., 2010b). In this

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regard, awareness of lameness as a problem is necessary for farmers to take action and implement prevention and control strategies (Ritter et al., 2017). Because veterinarians are often considered by farmers to be their main provider of health advice, their input on lameness management is often sought, yet herd veterinarians' ability to estimate lameness prevalence on farms they routinely work with has not been investigated.

Cow-based measures such as hock, knee, and neck injuries, as well as cleanliness scores, are also indicators of welfare, due to the pain they cause and their association with lameness and milk quality (Zurbrigg et al., 2005; Sant'Anna and Paranhos da Costa, 2011; Brenninkmeyer et al., 2013). These measures were also shown to have high prevalence in dairy herds (von Keyserlingk et al., 2012; Zaffino Heyerhoff et al., 2014; Jewell et al., 2019), but no evidence suggests that farmers and their veterinarians are aware of the prevalence of these welfare indicators on their own farms. Therefore, the objective of the present study was to compare farmers' and veterinarians' estimated herd prevalence of lameness, injuries, and cleanliness against the true prevalence of these issues on Québec dairy farms.

An observational cross-sectional study was conducted between February 2016 and July 2017, with the herd as the unit of interest. This study was approved by the animal care committee of the Université de Montréal (15-Rech-1786). The number of herds determined to be sufficient to quantify the agreement between estimated prevalence and the true herd prevalence, with a power of 80% and confidence of 95%, adjusted for clustering by veterinarian, was 81 (Abramson, 2011; Hulley et al., 2013). A convenience sample of clients of the bovine ambulatory clinic of the Faculté de médecine vétérinaire, Université de Montréal (Saint-Hyacinthe, QC, Canada) were enrolled in this study. All of the clinic's client herds (n = 135) were contacted to determine their willingness to participate in the study. Each herd that agreed to participate was visited by an animal health technician and a research veterinarian to complete the animal scoring. Prior to the farm visit, the farm owner and herd veterinarian had independently estimated, based on their recollection of their most recent interaction with the herd (biweekly or monthly herd visits), the prevalence of mild and severe lameness in the herd, as well as the prevalence of neck, knee, and hock injuries, and udder, leg, and flank cleanliness. All of the scoring methods, detailed as follows, were explained to farmers and veterinarians using picture charts before they had to estimate prevalence. Farmers and veterinarians were blinded to each other's estimations. Herds were then visited to evaluate cow-based measures (lameness, injuries, and cleanliness) in all lactating cows. The cows were all scored by the research veterinarian, and this point prevalence was considered as the true prevalence for the purposes of this study. The research technician scored a subsample of the cows to assess interobserver agreement.

In tiestall barns, lameness was evaluated with the animal standing, using the stall lameness scoring system described by Gibbons et al. (2014). Briefly, a cow was considered as lame if at least 2 of the lameness behaviors (weight shift, stand on edge, uneven weight, and uneven movement) were observed during a 1-min observation bout. In freestall barns, lameness was scored by looking at back arch, head bob, tracking-up, joint flexion, asymmetric gait, and reluctance to bear weight (Flower and Weary, 2006). A cow was considered as lame if its score was $\geq$3 on the 5-point lameness scale (1 and 2 = no limp; 3 to 5 = slight to severe limp). Cleanliness scoring was performed following the Cow Cleanliness Assessment (CBMRN, 2010), and cleanliness scores of 3 or higher for udder, legs, and flanks were considered as dirty. Swelling, skin lesion, or both on the neck, knees, or hocks were considered as injuries (Gibbons et al., 2012). The farmer, the herd veterinarian, the research veterinarian, and the technician were blinded to each other's results. Both the research veterinarian and the technician also scored the same 20 animals on 5 farms (total of 100 cows) for lameness, and later repeated their own scoring for the same 20 animals on those 5 farms (total of 100 cows), to allow for the estimation of intra- and interobserver agreement.

Statistical analyses were conducted using R version 3.6.0 (R Core Team, 2015). Intra- and interobserver agreement were assessed for lameness scoring using the kappa statistic (κ). The true prevalence and the estimated prevalence by farmers and veterinarians were first described for each cow-based measure. The association between estimated prevalence by the veterinarian and true prevalence was then evaluated using a linear mixed model, adjusted for clustering by veterinarian (random intercept; lme4 package). The association between estimated prevalence by the farmer and true prevalence was also evaluated but using a linear model (glm package). Type of housing and herd size were included as confounders if their inclusion in the model changed the estimate by more than 10% (Maldonado and Greenland, 1993). Two-way interactions with confounders were tested and considered as significant if $P < 0.05$. Normality and homoscedasticity of the residuals were assessed graphically using standardized residuals, and their fit was evaluated using outliers (residuals and student residuals), extreme (leverage), and influential data (Cook's distance and DFFITS). The results are presented for every 10% increase in true prevalence. The differences between estimated prevalence by farmers and veterinarians, and true prevalence, and the

differences between the estimated prevalence by the farmer and by the veterinarian were plotted (scatterplot). Descriptive statistics (median and range) were also calculated for these differences, for all herds and for herds with higher prevalence (highest quartile).

A total of 93 dairy herds were enrolled in this study and had, on average, 61.1 milking cows (SD = 29.5; median = 55.0), which was smaller than the average Québec herd (71 cows; $P < 0.01$; CDIC, 2017). Cows were housed in tiestall (90.3%; n = 84) and freestall barns (9.7%; n = 9), which did not differ from the overall housing types for Québec herds (86.1% tiestall, 13.9% freestall; $P = 0.32$; CDIC, 2017). All the herd veterinarians agreed to participate. The 10 participating veterinarians were each considered the primary herd veterinarian for 2 to 22 farms (median = 10.5). All veterinarians were involved with tiestall-housed herds (2 to 20 tiestall farms per veterinarian), and 4 veterinarians were also involved with freestall-housed herds (2 or 3 per veterinarian).

The interobserver agreement between the research veterinarian and animal health technician scoring was excellent ($\kappa = 0.84$), as was the intraobserver agreement on repeated scoring ($\kappa = 0.86$; MacLure and Willett, 1987). The true prevalence of cow-based measures observed on farms by the research team and the estimated prevalence by herd veterinarians and farm owners are presented in Table 1. The average prevalence of lameness was similar to previously published North American research (25 to 55%; Barker et al., 2010; von Keyserlingk et al., 2012; Bran et al., 2018). The average and maximum prevalences in the present study were, however, not as high as described recently in the province of Québec (Croyle, 2020). The prevalence of hock injuries was lower than what has previously been reported in freestall farm studies (29 to 81%; von Keyserlingk et al., 2012; Zaffino Heyerhoff et al., 2014; Jewell et al., 2019) but closer to the average found in a Canadian study, when including both freestall and tiestall farms (Croyle, 2020). The prevalence of knee and neck injuries in the present study was also different than what has been found in previous freestall farms studies (knee: 0 to 24%; neck: 9%; von Keyserlingk et al., 2012; Zaffino Heyerhoff et al., 2014; Jewell et al., 2019). Finally, the prevalence of cows with dirty udder, legs, and flanks fell within the wide range of previously published reports (udder: 4 to 71%; legs: 10 to 100%; flanks: 9 to 65%; de Vries et al., 2015; Croyle, 2020).

For all cow-based measurements, the prevalence estimated by the veterinarians increased as the true herd prevalence increased (Table 2). Similarly, the prevalence estimated by farm owners was positively correlated with the true prevalence, except for the prevalence of knee injuries (Table 2). This suggests that veterinarians and

Table 1. Descriptive statistics of the true herd prevalence, as observed by the research veterinarian on all milking cows of each herd, and the estimated prevalence by the farmers and veterinarians of 7 cow-based measures in 93 milking dairy herds in Québec, Canada[1]

	True herd prevalence (%)				Prevalence estimated by veterinarians (%)					Prevalence estimated by farmers (%)				
Measure	Mean	SD	Med	Range	Mean	SD	Med	Range	ρ (95% CI)	Mean	SD	Med	Range	ρ (95% CI)
Lameness	36.9	11.3	36.2	18.9–71.7	30.3	14.5	30	10–65	0.61 (0.47–0.73)	25.3	18.6	20	5–80	0.63 (0.49–0.74)
Neck injuries	16.7	16.3	14.3	0–64.7	15.1	11.8	10	0–60	0.58 (0.43–0.70)	14.7	15.0	10	0–95	0.43 (0.25–0.58)
Knee injuries	12.9	8.2	12.1	0–43.7	15.1	9.8	10	0–50	0.30 (0.10–0.48)	12.6	10.3	10	0–60	0.09 (−0.12–0.29)
Hock injuries	24.5	11.7	25.0	0–56.7	20.8	11.5	20	0–50	0.50 (0.33–0.64)	19.1	14.1	15	0–70	0.49 (0.32–0.63)
Dirty udder	14.4	10.9	12.5	0–55.0	16.7	12.3	15	0–50	0.60 (0.45–0.72)	16.7	16.4	10	0–90	0.70 (0.58–0.79)
Dirty legs	22.6	19.4	18.2	0–90.9	22.6	15.6	20	0–70	0.51 (0.34–0.65)	21.9	20.8	15	0–90	0.62 (0.47–0.73)
Dirty flanks	23.9	16.9	20.0	0–82.1	22.6	15.4	20	0–55	0.71 (0.60–0.80)	20.6	18.2	15	0–90	0.68 (0.55–0.78)

[1]Med = median. Pearson correlation coefficient (ρ) between each estimated prevalence and the herd prevalence is presented with 95% CI.

Table 2. Linear regression models between the true prevalence of 7 cow-based measures, as observed by the research veterinarian on all milking cows of each herd, and the prevalence of the same measures estimated by the herd veterinarian, adjusted for clustering by veterinarians (mixed model; random intercept), and estimated by the farmers in 93 dairy herds in Québec, Canada[1]

	Prevalence estimated by veterinarians			Prevalence estimated by farmers		
Item	Estimate	SE	*P*-value	Estimate	SE	*P*-value
Lameness						
Intercept	1.2	4.1	0.77	−15.2	5.4	0.01
For every 10% increase in true lameness prevalence	7.7	1.0	<0.01	11.0	1.4	<0.01
Neck injuries						
Intercept	7.4	1.6	<0.01	7.3	2.2	<0.01
For every 10% increase in true neck injury prevalence	4.4	0.6	<0.01	4.4	1.0	<0.01
Knee injuries						
Intercept	10.2	1.9	<0.01	13.2[2]	3.3	<0.01
For every 10% increase in true knee injury prevalence	3.8	1.2	<0.01	0.8[2]	1.5	0.62
Hock injuries						
Intercept	8.9	2.5	<0.01	2.9	3.2	0.37
For every 10% increase in true hock injury prevalence	4.8	0.9	<0.01	6.6	1.2	<0.01
Udder cleanliness						
Intercept	7.5	1.8	<0.01	−0.3	2.2	0.90
For every 10% increase in true dirty udder prevalence	6.5	0.9	<0.01	11.8	1.3	<0.01
Leg cleanliness						
Intercept	13.2[3]	2.1	<0.01	2.4[4]	2.7	0.02
For every 10% increase in true dirty leg prevalence	4.6[3]	0.8	<0.01	9.8[4]	0.9	<0.01
Flank cleanliness						
Intercept	7.7	2.2	<0.01	2.0	2.5	0.42
For every 10% increase in true dirty flank prevalence	6.4	0.6	<0.01	7.8	0.9	<0.01

[1]Type of housing and herd size were included in models as confounder if their inclusion changed the estimate by more than 10%. Models present the regression curve for every 10% increase in true herd prevalence. *P*-values were obtained by computation using Satterthwaite's method (lmerTest package, R version 3.6.0; R Core Team, 2015).

[2]Adjusted for type of housing and herd size.

[3]Adjusted for type of housing.

[4]Adjusted for type of housing and interaction between true prevalence and type of housing.

farmers enrolled in the present study were, on average, aware of the welfare issues on the farms.

Farmers generally underestimated the prevalence of lameness on their farm, with median true prevalence being 1.8 times higher than the farmers' estimate (range = 0.6 to 8.1). Interestingly, in herds with high lameness prevalence (>43%), the median true prevalence was only 1.2 times higher than the farmers' estimates (range 0.7 to 4.9). Although other studies have also found that farmers underestimated lameness prevalence on their farms (Espejo et al., 2006; Croyle, 2020), the farmers' estimations in the present study were closer to the actual prevalence. Farmers in the present study also slightly underestimated injury prevalence, with true prevalence being 1.1, 1.3, and 1.5 times higher than the neck, knee, and hock injury estimates of the farmers, respectively. In herds with high injury prevalence (neck >25%, knee >15%, and hock >33%), the true prevalence was 1.7, 2.2, and 1.5 times higher than the neck, knee, and hock injury estimates by the farmers, respectively. Cleanliness was most accurately estimated by the farmers, with the true prevalence being 1.0, 1.0, and 1.2 times higher than udder, legs, and flank cleanliness farmer estimates. Farmers overestimated the prevalence of dirty udder in herds with a higher prevalence of dirty udders (>20%; Figure 1E). Indeed, in herds with a high prevalence of dirty udders, legs, and flanks (udder >20%, legs >30%, and flank >35%), the true prevalence was 0.9, 1.2, and 1.2 times the udder, legs, and flank estimates by the farmers.

To our knowledge, this is the first time true herd prevalence has been compared with the estimated herd prevalence by the herd veterinarians. Based on our results, veterinarians underestimated the prevalence of these cow-based measurements, but generally less than farmers did (Figure 1). Indeed, the median true lameness prevalence was 1.2 times higher than veterinarians' estimates (range = 0.8 to 2.4). Compared with farmers, veterinarians were closer to the true prevalence values in herds with lower lameness prevalence (<40%), but the farmers and veterinarians similarly underestimated the prevalence in herds with higher lameness prevalence (>40%; Figure 1A). In herds with low injury or cleanliness prevalence, veterinarians overestimated, on average, the prevalence of the different cow-based measurements (intercept >0; Table 2). Veterinarians, as well as producers, slightly underestimated injury prevalence. Overall, the true prevalence was 1.1, 1.4, and 1.3 times higher than the neck, knee, and hock injury estimates by the veterinarians, respectively. In herds with

high injury prevalence (neck >25%, knee >15%, and hock >33%), the true prevalence was 1.6, 1.6, 1.4 times higher than the neck, knee, and hock injury estimates by the veterinarians, respectively. Veterinarians were also more accurately estimating the cleanliness indicators, with the true prevalence being 0.9, 0.8, and 1.0

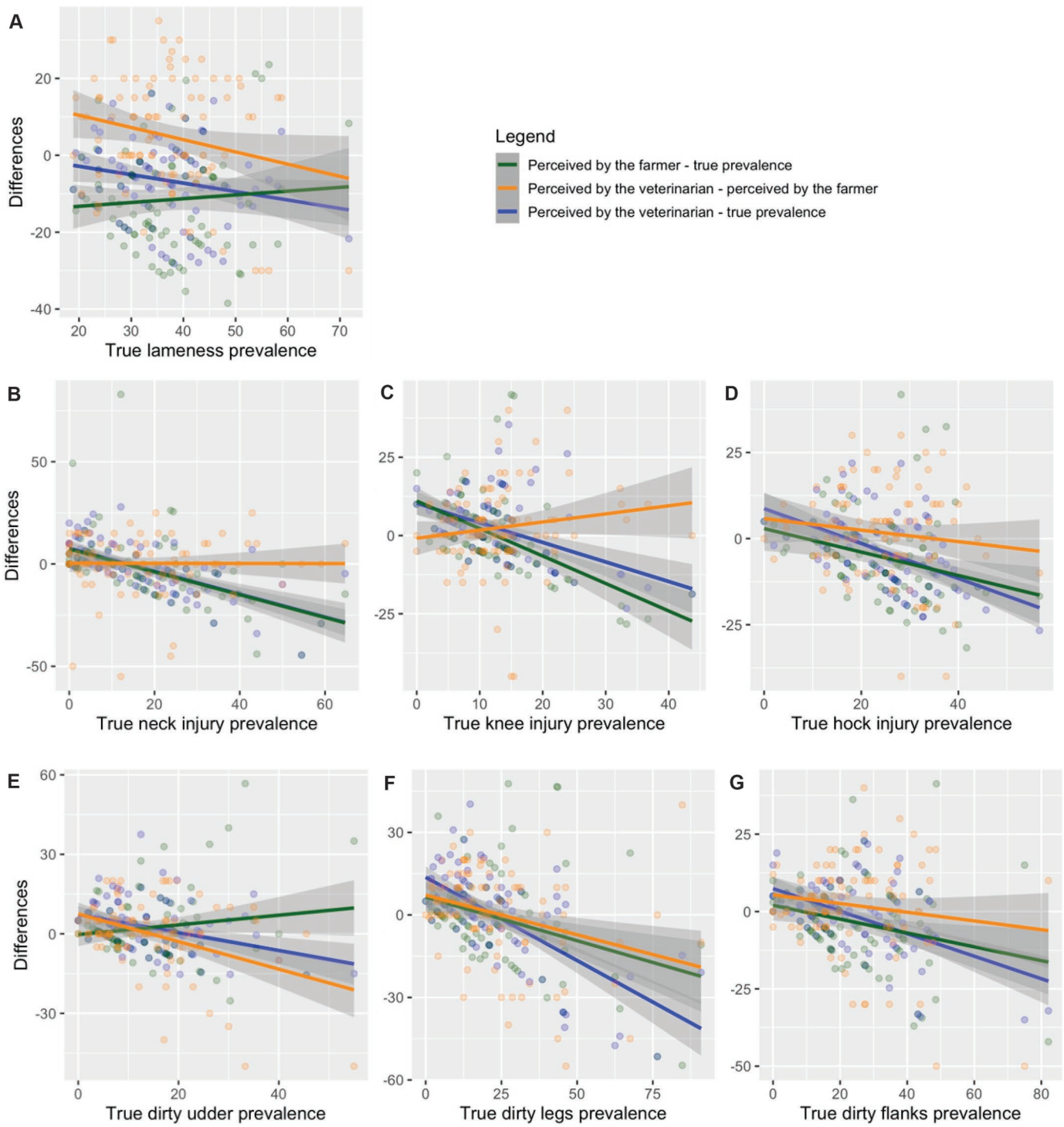

Figure 1. Differences between the prevalence estimated by veterinarian and farmers, and with the true prevalence, as observed by the research veterinarian on all milking cows of each herd, of (A) lameness, (B) neck injuries, (C) knee injuries, (D) hock injuries, (E) dirty udder, (F) dirty legs, and (G) dirty flanks, in 93 dairy herds in Québec, Canada.

times the udder, legs, and flank cleanliness estimates by the veterinarians. In herds with higher prevalence of dirty udders, legs, and flanks (udder >20%, legs >30%, and flank >35%), the true prevalence was 1.0, 1.5, and 1.2 times the udder, legs, and flank estimates by the farmers.

This study was conducted on a convenience sample of farms, all in the same region in Québec, Canada. As such, all of the veterinarians were working for the bovine ambulatory clinic of the Faculté de médecine vétérinaire, Université de Montréal (St-Hyacinthe, QC, Canada), which is a teaching center. No information was collected on the training each participant had on lameness, injuries, and dirtiness scoring before the start of the study. However, the scoring methods were explained in detail to all participants before they had to estimate prevalence. Because veterinarians involved in this study were working in a teaching center, it remains unclear whether their awareness about such scores was greater than for a veterinarian not working in a similar institution. Thus, it is unclear whether the results observed here can be generalized to other clinics, regions, and countries. This remains to be explored. Moreover, some of the farms in the present study had freestall housing, but most were tiestall barns, which could limit the ability to generalize these results to all types of farms. The type of housing, however, was retained as a confounder only in the models for leg cleanliness as estimated by veterinarians and farmers and the model for knee injuries as estimated by farmers, because only in these cases did it satisfy the statistical definition of confounding. This suggests that the discrepancy between the true prevalence and the estimated prevalence by veterinarians and farmers is likely similar on different types of farms, but more studies would be necessary to assess the generalizability of our results. As the veterinarians participating in this study are working with multiple herds, and some of the veterinarians were asked to estimate the prevalence of these cow-based measures in multiple herds, recall bias could have influenced their estimation. Although the time since last herd visit was not recorded at the time of scoring, it is unlikely that this bias influenced the results, as all the herds enrolled in the present study were frequently visited by their veterinarian (biweekly or monthly visits). Other factors, such as the ability of each participant to score lameness, injuries, and dirtiness, and the importance they gave to these conditions, could have influenced the present results but were not measured. Including such variables could have helped identify more precisely which farmers and veterinarians would benefit from targeted knowledge transfer in the future.

Farmers perceive the role of their veterinarians as to promote the health and welfare of their animals (Hall and Wapenaar, 2012), and underestimating the prevalence of a condition might minimize their effectiveness in this role, and consequently the priority that these issues receive on a farm (belief system; Garforth et al., 2004). The present study had no qualitative component, so the reasons for the discrepancies between the true herd prevalence and the estimated prevalence by the veterinarians and farmers were not explored. A qualitative exploration of the perception of lameness by these stakeholders in the dairy industry would enrich the findings of the present study (Olmos et al., 2018). It is also unclear whether a better awareness of true herd prevalence influences the measures taken by the veterinarians to address lameness on a specific farm, which could also be explored in the future. Studies have shown, however, that implementation of frequent lameness scoring by the veterinarian or a research team (with results communicated to the herd veterinarian) resulted in a general decrease of lameness prevalence (Main et al., 2012; Gundelach et al., 2013).

In conclusion, farmers and veterinarians enrolled in the present study generally underestimated the herd prevalence of lameness and injuries. Farmers with high herd lameness prevalence estimated their herd prevalence more accurately, which suggests good awareness in herds where lameness is an issue. Injuries were, however, less accurately estimated by veterinarians and farmers in herds with higher injury prevalence compared with herds with lower prevalence. This highlights an opportunity for better knowledge transfer in this area.

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